



Parul University
Faculty of Engineering and Technology
Parul Institute of Engineering and Technology
Computer Science & Engineering Department

Subject Name	Operating System	A.Y	2025-26
Subject Code	303105251	Semester	4th sem

Chapter-2

Sr No	Question	COs	B.T																								
1	Problem 1: Assume that you have the following processes all arriving at time 0: <table><tr><th>Process</th><th>Burst</th><th>Priority</th></tr><tr><td>P₁</td><td>8</td><td>4</td></tr><tr><td>P₂</td><td>6</td><td>1</td></tr><tr><td>P₃</td><td>1</td><td>2</td></tr><tr><td>P₄</td><td>9</td><td>2</td></tr><tr><td>P₅</td><td>3</td><td>3</td></tr></table> For each of the following CPU scheduling algorithms, determine the turnaround and wait times for each process as well as the average wait and turnaround times. First Come First Serve Shortest Job First Non-Preemptive Priority Scheduling Round Robin (assume a quantum of 1ms) Which of the four algorithms has the shortest wait time? Which has the fastest average turnaround time?	Process	Burst	Priority	P ₁	8	4	P ₂	6	1	P ₃	1	2	P ₄	9	2	P ₅	3	3	5	4						
Process	Burst	Priority																									
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P ₃	1	2																									
P ₄	9	2																									
P ₅	3	3																									
2	Problem 2: Let's change the assumptions and now assume that the jobs arrive at different times as following:- <table><tr><th>Process</th><th>Burst</th><th>Priority</th><th>Arrival Time</th></tr><tr><td>P₁</td><td>8</td><td>4</td><td>0</td></tr><tr><td>P₂</td><td>6</td><td>1</td><td>2</td></tr><tr><td>P₃</td><td>1</td><td>2</td><td>2</td></tr><tr><td>P₄</td><td>9</td><td>2</td><td>1</td></tr><tr><td>P₅</td><td>3</td><td>3</td><td>3</td></tr></table> Calculate the turnaround and wait times for the following algorithms: First Come First Serve Non-Preemptive Priority Scheduling Preemptive Priority Scheduling Round Robin (assume a quantum of 1ms)	Process	Burst	Priority	Arrival Time	P ₁	8	4	0	P ₂	6	1	2	P ₃	1	2	2	P ₄	9	2	1	P ₅	3	3	3	5	4
Process	Burst	Priority	Arrival Time																								
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P ₅	3	3	3																								
3	Problem 3: For the following page reference string: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 Calculate the page faults applying the following Page Replacement Algorithms for a memory with three frames: (i) Optimal (ii) LRU (iii) FIFO (iv) Second Chance	4	3																								

4	Problem 4: For the following page reference string: 6, 1, 2, 3, 5, 2, 1, 3, 2, 3, 6, 5, 4, 5, 2, 3, 1, 7, 0, 1 Calculate the page faults applying the following Page Replacement Algorithms for a memory with four frames: (i) Optimal (ii) LRU (iii) FIFO (iv) Second Chance	4	3
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Unit 3

Sr No	Question	COs	B.T
1	Give two differences between message passing and shared memory.	3	2
2	What is Inter-Process Communication (IPC)?	3	1
3	Define Critical Section.	3	1
4	What is a Race Condition?	3	1
5	Define Mutual Exclusion.	3	1
6	What is Peterson's Solution?	3	1
7	What is Inter-Process Communication (IPC)?	3	1
8	Explain the Producer–Consumer Problem using shared buffer.	3	2
9	Describe Semaphores. Explain wait() and signal() operations.	3	2
10	Explain Monitors with structure and working.	3	2
11	Explain Message Passing with send/receive primitives.	3	2
12	Explain the first Readers–Writers problem.	3	2
13	Explain the Dining Philosophers problem and starvation issue.	3	2
14	Explain the use of semaphores in classical IPC problems.	3	2
15	Compare Semaphores and Monitors.	3	3
16	List and explain the advantages of message passing over shared memory.	3	2
17	Explain the Producer–Consumer Problem using shared buffer.	3	2
18	Describe Semaphores. Explain wait() and signal() operations.	3	2
19	Explain Monitors with structure and working.	3	2
20	Explain Message Passing in detail. Compare direct vs indirect communication.	3	3
21	Explain Race Conditions and how semaphores eliminate them.	3	3
22	Explain various classical synchronization problems with solutions.	3	4
23	Explain Peterson's Solution in detail. Prove that it satisfies mutual exclusion and progress.	3	4
24	What is the Readers–Writers problem?	3	1
25	Define Deadlock.	3	1